METRIC

DOD-B-24504B(SH)
AMENDMENT-1
12 May 1980

MILITARY SPECIFICATION

BATTERY CELLS, STORAGE, SILVER-ZINC

ALKALINE TYPE (FOR DOLPHIN) (METRIC)

This amendment forms a part of Military Specification DOD-B-24504B(SH), dated 19 July 1978.

PAGE 1

2.1, under "SPECIFICATIONS, FEDERAL": Delete reference to "QQ-B-613", "QQ-B-626", "QQ-B-637", "QQ-B-639", "QQ-B-825", "QQ-C-576", "QQ-S-763", and "MMM-A-122".

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2.1, under "SPECIFICATIONS, MILITARY": Delete reference to "MIL-C-9084" and "MIL-R-9300", and add:

"MIL-P-46120 - Plastic Molding and Extrusion Material, Polysulfone."

2.1, under "STANDARD, FEDERAL": Delete reference to "FED-STD-595".

2.1, under "STANDARDS, MILITARY": Delete reference to "MIL-STD-1399, Section 101", and add:

"DOD-STD-1399, Section 532 - Interface Standard for Shipboard Systems, Cooling Water for Support of Electronic Equipment."

2.1, under "PUBLICATION, MILITARY": Delete reference to "NAVSHIPS 0901-LP-223-0000".

2.2, under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)": Add the following:

"A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
"B21 - Naval Brass Rod, Bar, and Shapes.
"B36 - Brass Plate, Sheet, Strip, and Rolled Bar.
"B121 - Leaded Brass Plate, Sheet Strip, and Rolled Bar.
"B152 - Copper, Sheet, Strip, Plate, and Rolled Bars.
"B187 - Copper Bus Bar, Rod, and Shapes."

FSC 6140

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PAGE 3

3.2.1, line 2: Delete "20-hour" and substitute "10-hour".

Table I: Delete and substitute:

"TABLE I. Electrical characteristics.

Discharge rate amperes (A)	Time hours (h)	Capacity (Ah)	Battery minimum voltage (V)	Minimum average 1/ cell voltage (V)	Minimum (cut-off) cell voltage (V)
3000	1	3000	182	1.29	1.13
1100	3	3300	214	1.45	1.31
750	5	3750	223	1.50	1.36
400	10	4000	230	1.50	1.39

1/ The "average voltage" is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 and 1 minute after applying a constant rate) and including the last recording. The interval of recording shall be not greater than h/5 hours where h is the time in hours of the rated capacity test. The time to the last recording shall be not greater than the rated time."

3.3.2, line 2: Delete "360,000 Ah" and substitute "240,000 Ah".

3.3.2, line 4: Delete "180,000 Ah" and substitute "120,000 Ah".

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3.3.4, line 2: Delete "3400 A" and substitute "3000 A".

3.3.8, lines 7 and 8: Delete ", with color No. 17038 in accordance with FED-STD-595".

3.4.3: Delete and substitute:

"3.4.3 Cell jar. Cell jars shall be molded out of polysulfone in accordance with MIL-P-46120. Cell jars shall be supported by uniformly-spaced pads cemented to the jar base so as to allow free passage of liquids under the cell jar. The minimum dielectric strength of an assembled jar shall be not less than 100 megohms when tested in accordance with 4.6.3.9."

3.4.4, first sentence: Delete and substitute: "The cell cover shall be molded out of the same material as the jar."

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3.4.4, line 5: Delete "QQ-S-763, class 304" and substitute "ASTM A276, type 304".

3.4.4, fifth sentence: Delete.

3.4.5: Delete and substitute:

"3.4.5 Cover-to-jar sealing. The cover-to-jar sealing shall be epoxy or cement."

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3.4.6.3, line 2: Delete "20-hour" and substitute "10-hour".

3.4.6.5, item (b), first sentence: Add ", or shall be used alone as the negative absorbent material".

3.4.6.5, item (c): Delete and substitute:

"(c) The positive and negative absorbent material shall be 0.08 to 0.25 mm thick. The material shall be a layer of nonwoven polyamide material from Pellon Corporation, or equal. The top shall be 5 to 6 mm over the top of the electrodes. This material shall be heat-sealed along the two vertical edges to form a bag around the electrode. If asbestos is used as the negative absorbent material, it shall be "U" wrapped around the electrode with vertical edges unsealed."

3.4.6.5, item (d), first sentence: Delete and substitute: "The main separator shall be Dupont Cellophane PUDO-193 (battery grade as is or treated) alone or in combination with Union Carbide Fibrous Sausage Casing (battery grade), or equal."

3.4.7, line 2: Delete "QQ-B-613, QQ-B-626, QQ-B-637, QQ-B-639, QQ-B-825, or QQ-C-576" and substitute "ASTM B21, B36, B121, B152, or B187".

PAGES 7 and 8

3.4.15 through 3.4.15.2: Delete and substitute:

"3.4.15 Negative electrode additive. The negative electrode additive shall be as specified in the contract or order (see 6.2.1). Mercury or mercury compounds are not allowed."

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3.7: Delete and substitute:

"3.7 Engineering services. Engineering services of the contractor shall be provided when specified (see 6.2.1) to:

(a) Assist the shipyard in preparing the cells for installation and installing them.
(b) Assist the ship's force in becoming familiar with the battery.

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(c) Take corrective action in case of unexpected problems.
(d) Examine the battery condition when requested.
(e) Participate in technical meetings with NAVSEA or the ship's force with regard to the battery and in locations other than the manufacturer's plant.

A maximum number of mandays allowed for the entire contract or life of the battery shall be as specified (see 6.2.1). Field engineering travel reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2)."

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4.5.2, line 1: Delete "written".

4.6.1.1: Delete.

4.6.3.2, line 3: Delete "200 A (20-hour rate), recharged, and then discharged at 3400 A" and substitute "400 A (10-hour rate), recharged, and then discharged at 3000 A".

4.6.3.2, line 5: Delete "20-hour" and substitute "10-hour".

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4.6.3.7, line 4: Delete "3400 A" and substitute "3000 A".

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4.6.3.9: Delete and substitute:

"4.6.3.9 Dielectric test. Each sample cell shall be submitted to the dielectric measurement in accordance with figure 5. The dielectric strength of the sample cells shall meet the requirements of 3.4.3."

4.6.3.11, line 4: Delete "MIL-STD-1399, section 101" and substitute "DOD-STD-1399, Section 532".

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5.3.1.1, item (f): Delete.

6.2.1, item (b): Delete.

6.2.1, item (c): Delete and substitute:

"(c) Identification and amount of negative electrode additive (see 3.4.15)."

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6.2.1, item (i): Delete "total" and substitute "maximum".

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6.4 through 6.4.5: Delete and substitute:

"6.4 Guarantee. The guarantee of the cells when series-connected as a battery shall be as specified in 6.4.1 through 6.4.4.

"6.4.1 Nothing set forth in this guarantee provision shall limit any rights afforded to the Government by the inspection clause, relating to latent defects, fraud, or gross mistakes that amount to fraud. This guarantee applies notwithstanding inspection and acceptance by the Government of supplies furnished under the contract or any provision of this contract concerning the conclusiveness thereof. The contractor guarantees the performance of the battery in terms of service operating conditions (months of service or total Ah output), when the battery is operated and maintained in accordance with the service manual. The battery shall be guaranteed for 18 months or to deliver 240,000 Ah, whichever occurs first, before the battery capacity drops to 60 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification. The period of the guarantee shall begin:

(a) Date the cells are filled and formed in accordance with the manufacturer's instructions, or
(b) 24 months after acceptance of the sample cell test report in accordance with 4.6.3, whichever occurs first.

"6.4.2 In the event of a failure of the battery of at least five cells, as specified in 6.4.3 the contractor shall:

(a) When battery capacity drops below 80 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification within the first 12 months or 120,000 Ah, whichever occurs first, the contractor shall replace the total battery, with a new unused battery manufactured in accordance with this specification at no cost to the Government.
(b) After the period specified in (a) and prior to the end of the guarantee life of 18 months or 240,000 Ah, whichever occurs first, the contractor shall indemnify the Government on a pro-rata basis (proportionately) for months of service not obtained or total Ah output not obtained, whichever is smaller. (If P being the purchase price of the battery, N the number of months not obtained, and Q the total Ah output not obtained, the smaller of the two numbers $P \frac{N}{18}$ and $P \frac{Q}{240,000}$ shall be used.)

"6.4.3 The battery shall be considered failed in the following cases:

(a) When grounds develop which cannot be corrected by the contractor to meet the requirements of the service manual. Corrective action by the contractor shall be at his own expense.

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(b) When the battery minimum voltage specified in table I limits battery capacity to a percentage specified in 6.4.2 and cannot be restored by contractor corrective action which will be at his own expense.
(c) When 30 or more cells fail, with any cell considered failed in the following cases:
(1) The cell is internally shorted.
(2) The cell limits the battery capacity to a percentage specified in 6.4.2 and has to be removed in order to raise the overall battery capacity.
(3) The cell develops leaks through cracks, thus causing grounds in the system.

"6.4.4 The same guarantee requirements of life, total Ah output, minimum capacity and indemnification specified in 6.4.1, 6.4.2, and 6.4.3 shall apply to each spare cell individually, whether wet or dry. Nothing in 6.4 diminishes the Government's rights contained in any other provision of the contract, including but not limited to inspection."

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Add the new attached figure 5 on page 24A.

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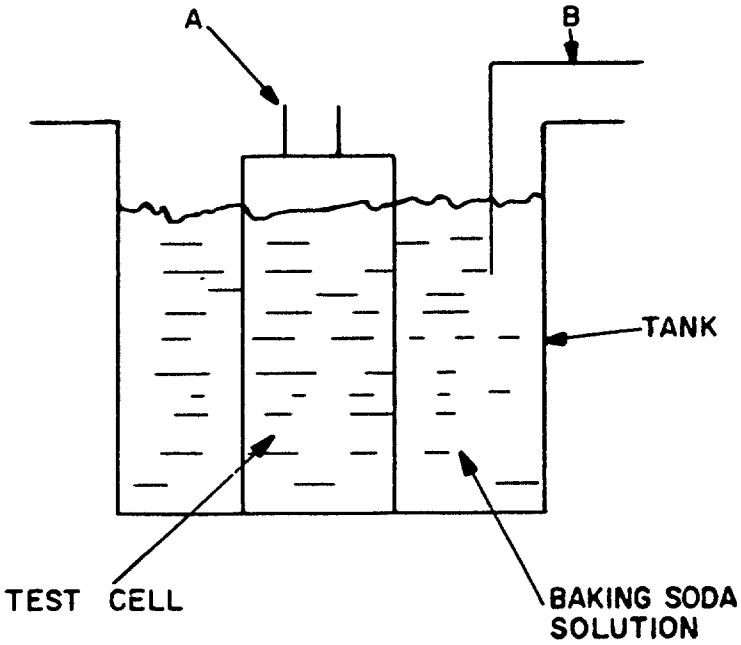
30.8.5, item (ss): Delete.

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Example G, under "HAZARDS": Delete "815 Mercury".

Preparing activity:
Navy - SH
(Project 6140-N523)

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SH 11540

Measure dielectric strength at points A and B using a General Radio megohmmeter type 1864, or equal.

FIGURE 5. Cell dielectric measurement.

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